

2.10.5 [6 points] Analyse the impact of skipping time steps during the generation process (generation of images with reverse diffusion process). Provide your hypothesis of how this might influence the final output. Write if the results met your expectations and why. (150 words max)

Hypothesis. The DDPM reverse mean uses coefficients calibrated to the assumed noise level t . Skipping breaks this assumption. For skip $[5, 75]$, the low-noise tail steps remove almost no noise (small βt), so the chain cannot clean up a still-noisy x_{76} . For skip $[130, 200]$, the model treats pure noise as level 129 and over-interprets it as if it already contained signal.

Results. Skip $[5, 75]$ outputs at $t = 0$ remain **essentially noise**: structure barely emerges, since jumping from $t = 76$ to $t = 4$ strips only a negligible amount of noise. Skip $[130, 200]$ produces **visible high-contrast patterns** but with artefacts and no clear garment silhouettes: the network hallucinates structure from a too-noisy starting state.

Comparison. Both regimes degrade samples, as predicted. Skip $[5, 75]$ was harsher than expected (no silhouettes at all). Skip $[130, 200]$ matched our qualitative prediction of structured-but-distorted outputs.

2.10.6 [8 points] The training of Denoising Diffusion Probabilistic Models often replaces the loss with a simplification of the loss, DDPM (see [the original](#) paper). Explain how this simplified objective relates to and why it is considered a valid alternative. (250 words max)

*Note: When you make a statement, you need to ensure that it is supported with an academic resource. Even though you use ChatGPT for understanding the theory, you need to support the claims with **peer-reviewed academic** resources. Standard referencing practices apply. Use IEEE referencing style. Not referencing a used source will be penalised.*

We train diffusion models by minimising a variational upper bound on the negative log-likelihood, $LVLB = \mathbb{E}_q[LT + \sum L_{t-1} + L_0]$, where $L_{t-1} = DKL(q(x_{t-1}|x_t, x_0) \parallel p_\theta(x_{t-1}|x_t))$ [1]. Fixing the reverse variance and adopting the ε -reparametrisation $\mu_\theta(x_t, t) = (1/\sqrt{\alpha_t})(x_t - \beta_t/\sqrt{(1-\bar{\alpha}_t)} \cdot \varepsilon_\theta(x_t, t))$ collapses each L_{t-1} into a noise-prediction MSE weighted by $w_t = \beta_t^2 / (2\sigma^2 \alpha_t(1-\bar{\alpha}_t))$ (Ho et al., Eq. 12 [1]).

The simplified objective $L_{simple} = \mathbb{E}_{t, x_0, \varepsilon}[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2]$ of Ho et al. (Eq. 14 [1]) replaces w_t with unit weights and samples t uniformly. Arbitrarily reweighting the per-step KL breaks the Jensen step yielding $LVLB$, so **L_{simple} is a surrogate, not an upper bound on $-\log p_\theta(x_0)$** [1].

We consider it a valid alternative on two grounds. **Empirically**, Ho et al. report on CIFAR-10 a marginal NLL increase (**3.70 $\rightarrow$ 3.75 bits/dim**) but a dramatic FID improvement (**13.51 $\rightarrow$ 3.17**) under L_{simple} compared to $LVLB$ [1]. **Theoretically**, the optimal MSE noise predictor coincides with $\mathbb{E}[\varepsilon|x_t]$, proportional to the score $\nabla \log q_t(x_t)$; this links L_{simple} to denoising score matching, established by Hyvärinen [2] and Vincent [3], and generalised across multiple noise scales by Song and Ermon [4]. Kingma et al. later showed that a closely related denoising loss, weighted by signal-to-noise ratio in continuous time, recovers a tight variational bound, confirming that the family of objectives is principled when correctly weighted [5].

Link to ChatGPT conversation: [link](#).

References

- [1] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 6840-6851.
- [2] A. Hyvärinen, “Estimation of non-normalized statistical models by score matching,” Journal of Machine Learning Research, vol. 6, pp. 695-709, Apr. 2005.
- [3] P. Vincent, “A connection between score matching and denoising autoencoders,” Neural Computation, vol. 23, no. 7, pp. 1661-1674, Jul. 2011.
- [4] Y. Song and S. Ermon, “Generative modeling by estimating gradients of the data distribution,” in Advances in Neural Information Processing Systems, vol. 32, 2019.
- [5] D. P. Kingma, T. Salimans, B. Poole, and J. Ho, “Variational diffusion models,” in Advances in Neural Information Processing Systems, vol. 34, 2021.